FINAL PROJECT REPORT

Sentiment Analysis for University Brand using Social Media Data

Course: Social Mining
Submitted by: *Vedhika Reddy Challa - 000802364*

1. Introduction

In recent years, the digital transformation of communication has significantly influenced how individuals share their experiences and opinions. Higher education institutions, including universities, are no exception. Public perception of a university now extends far beyond physical interactions; it is shaped largely by online discussions, reviews, and posts shared across multiple social media platforms.

Students today openly talk about their academic experiences, faculty interactions, campus life, infrastructure, fee-related issues, and administrative support systems on social networking sites. These platforms serve as a rich repository of opinions that reflect the true sentiment of the student community.

This shift has made Sentiment Analysis—the computational study of opinions, emotions, and attitudes expressed in text—an essential tool for understanding a university's brand perception. Using Social Mining techniques, universities can analyze thousands of reviews quickly and make informed decisions to improve their reputation and service quality.

This project aims to apply sentiment analysis on real student reviews to understand how students perceive the university's brand, identify areas of praise, uncover recurring complaints, and interpret the overall public sentiment.

2. Problem Statement

Higher education institutions care deeply about how they are perceived by students and society. A university's brand influences:

- Student admissions
- Alumni support
- Funding opportunities
- Public trust and reputation

However, there is currently no automated system for the university to evaluate real-time online sentiment from social networks.

Key problem:

How can the university effectively analyze and interpret thousands of social media posts to understand its brand reputation in real time?

The challenge arises because:

- The volume of data is large
- Human analysis is slow and inconsistent
- Social media language is informal, emotional, and noisy
- Opinions fluctuate rapidly

Thus, a computational approach is needed to automatically mine opinions and classify them as positive, negative, or neutral.

3. Objectives of the Project

The project aims to provide an end-to-end sentiment analysis pipeline.

Primary Objective

To build a machine learning model capable of classifying student reviews as positive or negative, thereby helping the university measure its brand perception.

Specific Objectives

1. To clean and preprocess raw textual data using NLP techniques.
2. To convert the cleaned text into numerical features using TF-IDF.
3. To train a classification model using a machine learning algorithm (Multinomial Naive Bayes).
4. To evaluate the performance using accuracy, precision, recall, and F1-score.
5. To visualize the sentiment distribution using graphs and word clouds.
6. To interpret the results in the context of student satisfaction and university branding.

4. Literature Review

Sentiment Analysis has been extensively researched as part of opinion mining and text analytics.

4.1 Importance of Sentiment Analysis

Sentiment analysis allows institutions and businesses to gain insights from unstructured text. It is widely used in:

- Customer reviews
- Social media monitoring
- Product feedback systems
- Political opinion tracking
- Reputation management

4.2 Approaches to Sentiment Analysis

(a) Lexicon-Based Approaches

These depend on predefined dictionaries of positive and negative words.

- Pros: Easy to use
- Cons: Cannot detect sarcasm, context, or domain-specific meanings

(b) Machine Learning Approaches

Algorithms learn from labeled data.
Common models:

- Naive Bayes
- Logistic Regression
- Support Vector Machines

These models require feature extraction methods such as TF-IDF.

(c) Deep Learning Approaches

Use neural networks like:

- LSTMs
- GRUs
- BERT-based transformers

They achieve higher accuracy but require more resources.

For a student project with a structured dataset, Multinomial Naive Bayes is effective and computationally efficient.

5. Dataset Description

The dataset consists of student reviews collected from online sources. Each review is labeled as either positive (1) or negative (-1).

Dataset Summary (from your PDF):

- Total samples: 9,426
- Positive entries: 7,817
- Negative entries: 1,609

This shows that:

- The dataset is highly imbalanced, with a greater number of positive reviews.
- Students are generally more inclined to share positive experiences.

The dataset includes a wide variety of reviews covering topics such as academics, placements, campus facilities, hostel experience, extracurricular activities, administrative response, and fee structure.

6. Methodology

The methodology is divided into the following phases.

6.1 Data Preprocessing

The raw text contains noise such as:

- Uppercase and lowercase mix

- Special characters
- URLs
- Stop words
- Misspellings
- Extra spaces

The following preprocessing steps were applied:

1. Lowercasing: Converts all text to lowercase to ensure uniformity.
2. Removing non-alphabetic characters: Removes numbers, symbols, and punctuation.
3. Tokenization: Splits text into individual words.
4. Stop word removal: Eliminates common words like *"is", "the", "and"* which carry little meaning.
5. Lemmatization: Converts words to their root form.
6. Creation of 'clean_text' column: Stores cleaned reviews used for model training.

6.2 Feature Extraction

We used TF-IDF Vectorizer (max_features=5000), a standard technique to convert text to numerical vectors.

TF-IDF helps the model distinguish important words by giving:

- Higher weight to meaningful, unique words
- Lower weight to overly common words

Common high TF-IDF words in the dataset include:

- *“gained”, “experience”, “supportive”, “faculty”, “building”, “friends”, “time”, “offer”, “advise”*

6.3 Model Selection and Training

We used Multinomial Naive Bayes, a popular supervised classification algorithm for text data.

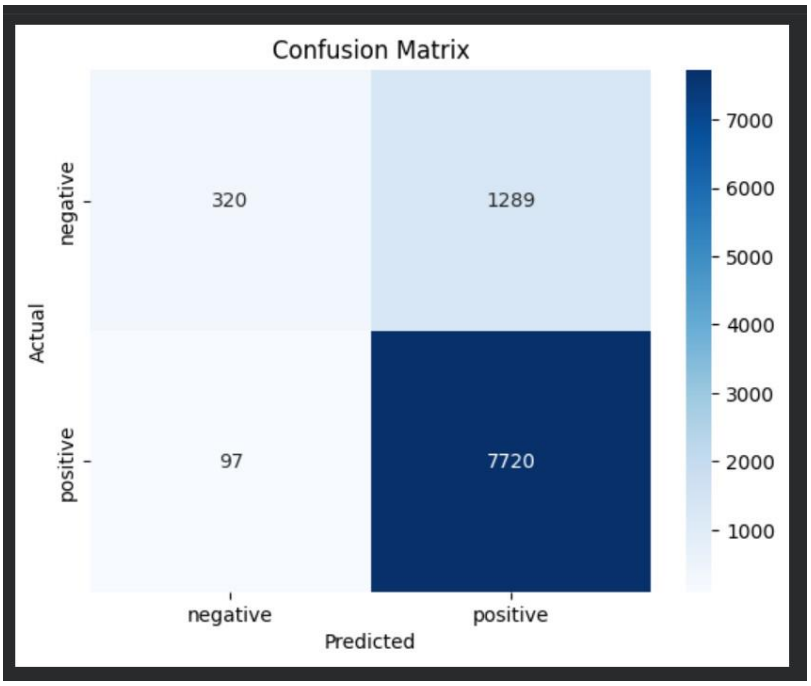
Reason for selecting Naive Bayes:

- Works well on document-term matrices
- Fast training time
- Performs well with TF-IDF features
- Suitable for large datasets

The dataset was split into:

- Training set: 80%
- Test set: 20%

The model was trained using TF-IDF vectors as input features.



6.4 Evaluation Metrics

From the output:

Accuracy: 85.3%

Classification Report

Metric	Negative	Positive
Precision	0.77	0.86
Recall	0.20	0.99
F1-score	0.32	0.92
Support	1609	7817

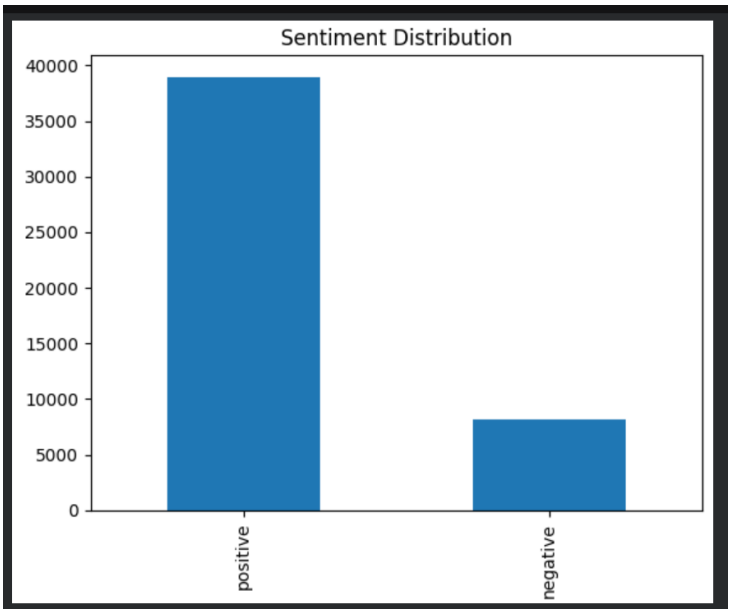
Overall Averages

- Macro Avg F1: 0.62
- Weighted Avg F1: 0.81

Sentiment Distribution (Dataset Level):

- Positive: 83%
- Negative: 17%

These values highlight that the model performs extremely well for positive reviews but struggles with the smaller negative class.



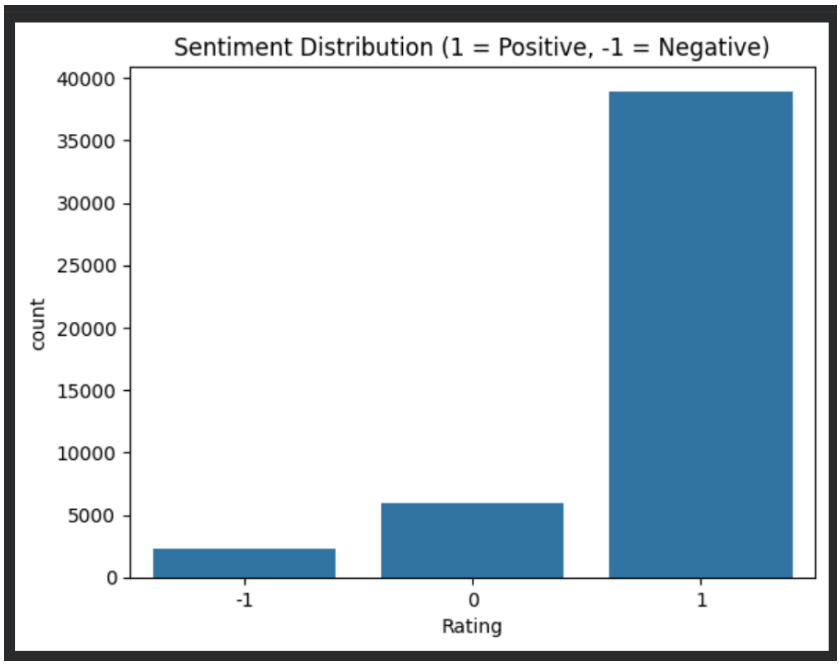
7. Results and Analysis

7.1 Positive Sentiment Analysis

A majority of students expressed positive experiences, praising:

- Campus life
- Teaching quality and faculty support
- Infrastructure and facilities
- Availability of extracurricular activities
- Opportunities for personal and professional growth

Word cloud and TF-IDF weights show strong presence of positive terms such as *“good”, “supportive”, “experience”, “friends”, “building”, “loved”.*



7.2 Negative Sentiment Analysis

Although fewer, the negative reviews highlight:

- Administrative delays
- Hostel and food issues
- Unclear or slow communication from staff
- Expensive fees
- Maintenance concerns

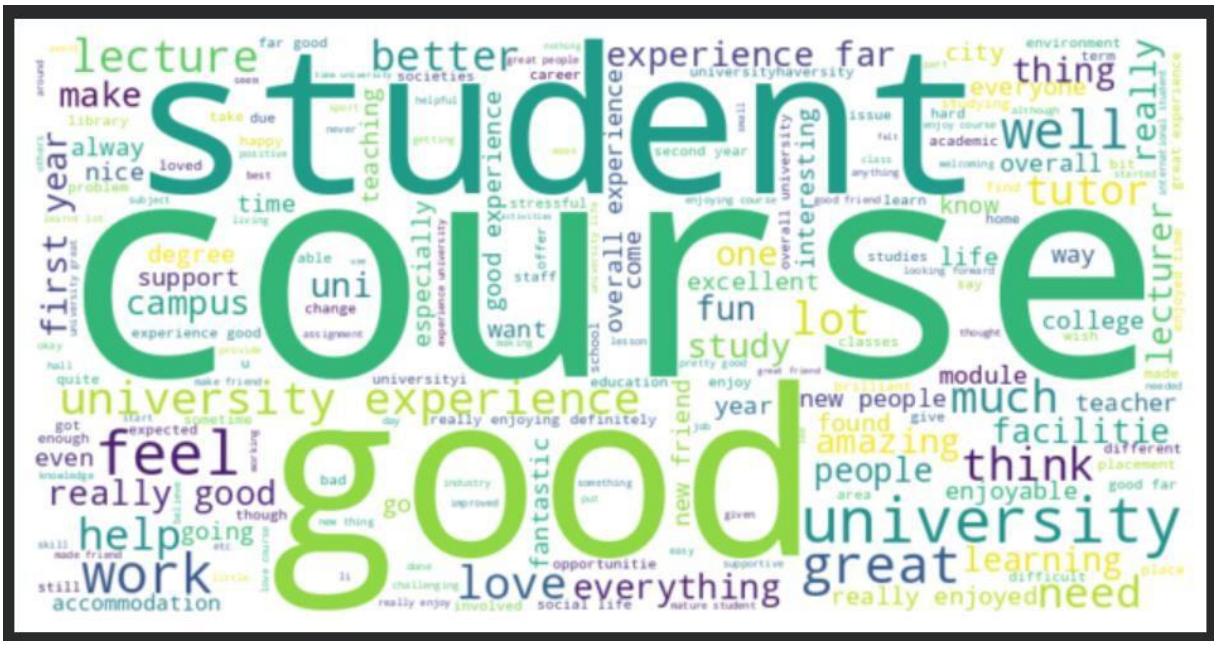
These insights reveal areas where the university can improve student satisfaction.

7.3 Performance Interpretation

- The model identifies **positive sentiment very accurately** (recall: 0.99).
- However, it misclassifies many negative reviews due to:
  - Class imbalance
  - Short or ambiguous text

- Lack of strong negative keywords

This is expected in sentiment analysis of imbalanced data.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Despite limitations, the model achieves good overall accuracy and provides meaningful insights.

9. Conclusion

This project successfully demonstrates how Social Mining and Sentiment Analysis can be applied to evaluate a university's brand reputation. By analyzing social media posts, we gained insights into public perception, strengths, and areas needing improvement.

The project helped us understand:

- The end-to-end pipeline of text mining
- Importance of data preprocessing
- How machine learning models classify sentiment
- How visualizations help interpret social data

Overall, sentiment analysis proves to be a powerful, scalable tool for educational institutions to understand their image and take informed decisions to enhance student and public satisfaction.

10. References

- Bing Liu, "Sentiment Analysis and Opinion Mining"
- Course lecture slides and Social Mining materials
- Research papers on sentiment classification
- Online resources related to TF-IDF and NLP preprocessing